

Module 3 Contribution Journal

ITAI 1371 – Machine Learning

Neural Crew

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For the Module 3 lab, I contributed by completing the Machine Learning Workflow and Types of Learning notebook along with my group. Since our group members are all completing the same lab and working through the same material, my contribution was mainly making sure I understood the notebook, completed the required sections correctly, and was able to compare my work with the group if anyone had questions or needed help checking something.

I worked through the notebook in Google Colab and ran the code cells as I went instead of just running everything at once. I am still learning Python and machine learning, so I tried to slow down and look at what each section was doing. I paid attention to the outputs and the way the code used the dataset to train and test the model. This helped me better understand the workflow instead of only focusing on getting the assignment finished.

One part of the lab I completed was the section about the different types of machine learning. I answered the questions that asked whether certain examples were supervised learning, unsupervised learning, or reinforcement learning. I reviewed the differences between them before answering because it can be easy to mix them up at first. Supervised learning uses labeled data to make predictions, unsupervised learning looks for patterns in data without labels, and reinforcement learning involves learning through rewards and consequences. Completing that part helped me understand why different types of machine learning are used for different problems.

I also worked with the Wine dataset and the classification model in the notebook. In one section, we were able to choose three features and see how they affected the model's accuracy. I looked at the results and compared the accuracy from the selected features to the accuracy from the original model. This showed me that the features used in a machine-learning model can make a difference in how well it performs. A model does not always improve just because changes are made. Sometimes a different feature selection can lower the accuracy, which is why testing and comparing results is important.

Another part of my contribution was completing the written reflection and analysis questions in the notebook. I explained the three types of machine learning in my own words and wrote about the model results. I also thought about why Logistic Regression may have performed well for the dataset and what could possibly be changed or tested in the future to improve the model.

For my real-world machine-learning example, I focused on the cinema and movie-theater industry. I thought about how theaters or movie companies could use data from ticket sales over

time to see what types of movies people in a particular city are most likely to watch. For example, a theater could look at information such as movie genre, rating, release date, ticket sales, showtimes, season of the year, and location. The data could show whether people in one city tend to watch more horror movies, action movies, family movies, comedies, or R-rated movies compared to people in another city.

Machine learning could use that information to help theaters decide which kinds of movies may be more successful in certain locations. For example, if ticket-sale data shows that horror movies and R-rated thrillers perform especially well in one city during October, a theater could use that information when deciding how many showtimes to schedule for similar movies. It could also help movie studios and theaters with advertising, promotions, and deciding which movies should have a wider release in certain areas. I thought this was a good example because it shows how machine learning can use past customer data to make predictions about what people may want to see in the future.

I also made sure my notebook was completed and saved correctly. I checked my assessment answers, reviewed the written reflection, and made sure the notebook was ready to be shared with the group. Since we are all doing the same work, it is still helpful to compare that everyone completed the requirements and that nobody missed an important section. This also gives us a chance to discuss parts of the assignment that may have been confusing.

Overall, I feel that I contributed by carefully completing the Module 3 notebook, checking my answers, reviewing the machine-learning concepts, working with the Wine classification model, and completing the written reflection. Even though our group members all complete the same lab, I believe my contribution is making sure that I do my part thoroughly and understand the material well enough to participate in group discussions and help check the final work. I am still new to machine learning and Python, but I feel more comfortable than I did at the beginning of the course.